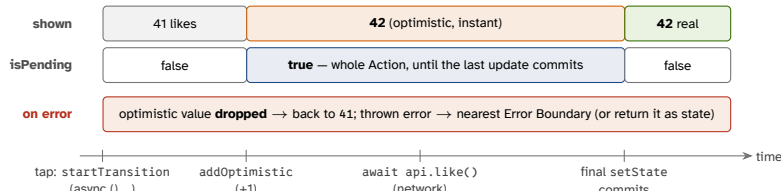


In one line: React 19 (Dec 2024) makes async updates first-class: an **Action** is a function run in a **transition**, and React tracks its **pending**, **optimistic** and **error** state (`useActionState`, `useOptimistic`). 19.2 (Oct 2025) adds `<Activity>` and `useEffectEvent`; **React Compiler 1.0** (Oct 2025) memoises components and hooks **at build time**. RN: React 19.0 in **0.78**, 19.1 in **0.80**, 19.2 in **0.83**.

An Action with an optimistic update — “Like” tapped



React Compiler: what the Babel plugin emits

function List({items}) {
 const s = sort(items);
 return <Rows data={s}/>;
}

build →

const \$ = _c(4); // cache slots
if (\$[0] !== items) {
 s = sort(items); \$[0]=items; \$[1]=s;
} else s = \$[1]; // same for JSX

HIR (control-flow graph) + data-flow and mutability analysis decide each value's inputs

\$[0] \$[1] \$[2] \$[3]

one cache per component instance (useMemoCache); memoises values, callbacks and JSX — also after an early return

breaks a **Rule of React** (mutates props, reads a ref in render...)? → that function is **skipped silently**, still runs uncompiled

React 19 — Actions and friends

- Action** = function run in a transition (`startTransition(async () => {...})` or `useTransition`). `isPending` stays true until the *final* update commits. State set *after* an await is not in the transition — wrap it in `startTransition` again.
- `useActionState(action, initial) → [state, dispatch, isPending]`; `action(prevState, payload)` returns the next state. Dispatches are **queued** and run in order. Renamed from canary `useFormState`.
- `useOptimistic(value, reducer?) → [shown, addOptimistic]`. Call `addOptimistic` *inside* an Action; when it settles React falls back to value — no manual rollback.
- `use(promise | context): suspends` until resolved (Suspense + Error Boundary); the one API allowed in `if/loops`. The promise must be *stable* (from a parent, cache or library) — not created in render.
- ref is a plain prop** of function components: `forwardRef` no longer needed, to be deprecated. Ref callbacks may **return a cleanup**. `<Ctx value={v}>` replaces `<Ctx.Provider>`.
- Web-only (react-dom): `<form action>`, `useFormStatus`. RN: `dispatch` from `onPress`.

React 19.2 (RN 0.83) and 19.3

- `<Activity mode="visible"|"hidden">`: `hidden` = children hidden, **effects unmounted** (cleanups run), updates deferred to idle, **state kept**. Pre-render the next tab; keep a back screen's state without its subscriptions.
- `useEffectEvent(fn)`: an “event” called from an Effect that reads the *latest* props/state but is **not reactive** — never in deps, only called from Effects of the same component. Needs `eslint-plugin-react-hooks@latest`.
- 19.2 also: **Performance Tracks** (Scheduler, Components) in the profiler. 19.3 (9 Sep 2026): `<ViewTransition>`, `addTransitionType`, **Fragment refs**.

Example — an Action in RN (19.x)

```
function LikeButton({ post }) {
  const [shown, addOptimistic] = useOptimistic(post.likes);
  const [error, like, isPending] = useActionState(
    async (_prev, id) => {
      addOptimistic(n => n + 1); // instant UI
      try { await api.like(id); return null; }
      catch (e) { return 'Could not like'; } // UI reverts
    }, null);
  return (
    <Pressable disabled={isPending}
      onPress={() => startTransition(() => like(post.id))}>
      <Text>{shown} likes {error ?? ''}</Text>
    </Pressable>;
  )
}
```

Does the compiler end useMemo/useCallback?

For **new** code: mostly — write plain code. Keep them as **escape hatches** where identity is a *contract* (an effect dep that must not change). Old ones: leave, or remove with tests — removing changes the compiled output. Memo is **per component**: an expensive function called in 3 components still runs 3 times.

React Compiler 1.0

- `babel-plugin-react-compiler` (first Babel plugin) lowers each component/hook to an **HIR**, infers each value's inputs, emits a per-instance cache (`_c(n)`). Granular — even **after conditional returns**, which manual hooks cannot do.
- `compilationMode: 'infer'` (default: PascalCase + JSX/hooks, or `use* + hooks`) · ‘annotation’ (only “use memo”) · ‘syntax’ (Flow) · ‘all’. “use no memo” opts one function out (debugging).
- Assumes the Rules of React**: render is pure and idempotent, props/state/hook results immutable, hooks at top level, no `ref.current` in render. Unsure → **skip** that function (panicThreshold: ‘none’) — never a build break.
- Lint**: `eslint-plugin-react-hooks` (v6+, flat config) recommended carries the compiler rules (`set-state-in-render`, `set-state-in-effect`, `refs`, ...). Fix lint first: a flagged component is a skipped one.
- React 17/18: target + `react-compiler-runtime`. Verify: a **sparkle “Memo” badge** in React DevTools. **RN**: supported; **Expo SDK 54+** on by default; bare RN adds the Babel plugin. Meta Quest Store: loads up to 12% faster.

Example — before / after the compiler

```
// BEFORE: manual, and still broken
const Row = memo(({ item, onPick }) => ...);
function List({ items, onPick }) {
  const sorted = useMemo(() => sort(items), [items]);
  const pick = useCallback(id => onPick(id), [onPick]);
  return sorted.map(i =>
    <Row key={i.id} item={i} onPick={() => pick(i.id)} />;
  )
} // inline arrow = new prop each render: memo defeated
// AFTER: plain code; compiler caches sorted, the arrows
function List({ items, onPick }) {
  const sorted = sort(items);
  return sorted.map(i =>
    <Row key={i.id} item={i} onPick={() => onPick(i.id)} />;
  )
}
```

Traps

- ! `use(fetchUser())` in render: new promise each render → suspends forever.
- ! `addOptimistic` outside an Action: warning, never shown.
- ! Code that *relied* on memo identity (effect deps) can change behaviour — bisect with “use no memo”.
- ! Activity hidden ≠ unmounted: state stays, effects stop.

Remember

Action = transition + pending + optimistic + error. Compiler = memo by data-flow, if you obey the rules.